EXPERIMENT : 17

AIM:

Experiment to understand Linear Regression for a given data set.

DESCRIPTION:

Understand the Linear regression for the dataset given

CODE:

```
import numpy as np
import pandas as pd

# Read dataset
df = pd.read_csv('/content/salary.csv')

print(df.head())
df.info()

# Remove missing values
df.dropna(inplace=True)
df.info()

# Display statistics
print(df.describe())

# Features and label
features = df.iloc[:, [0]].values
label = df.iloc[:, [1]].values

# Split dataset
from sklearn.model_selection import train_test_split

x_train, x_test, y_train, y_test = train_test_split(
    features, label, test_size=0.2, random_state=42
)
```

```
# Create and train model
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(x_train, y_train)

# Model accuracy
print("Training Score:", model.score(x_train, y_train))
print("Testing Score:", model.score(x_test, y_test))

# Coefficient and intercept
print("Coefficient:", model.coef_)
print("Intercept:", model.intercept_)

# Save model
import pickle

pickle.dump(model, open('SalaryPred.model', 'wb'))

# Load model
model = pickle.load(open('SalaryPred.model', 'rb'))

# Predict salary
yr_of_exp = float(input("Enter Years of Experience: "))

yr_of_exp_NP = np.array([[yr_of_exp]])

Salary = model.predict(yr_of_exp_NP)

print("Estimated Salary for {} years of experience is {:.2f}".format(
    yr_of_exp, Salary[0][0]
))
```

OUTPUT:

```
  Unnamed: 0  work_year experience_level employment_type  \
0          0       2020              MI              FT
1          1       2020              SE              FT
2          2       2020              SE              FT
3          3       2020              MI              FT
4          4       2020              SE              FT

                  job_title  salary salary_currency  salary_in_usd  \
0             Data Scientist   70000             EUR          79833
1  Machine Learning Scientist  260000             USD         260000
2          Big Data Engineer   85000             GBP         109024
3       Product Data Analyst   20000             USD          20000
4   Machine Learning Engineer  150000             USD         150000

  employee_residence  remote_ratio company_location company_size
0                 DE             0               DE            L
1                 JP             0               JP            S
2                 GB            50               GB            M
3                 HN             0               HN            S
4                 US            50               US            L
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 607 entries, 0 to 606
Data columns (total 12 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   Unnamed: 0          607 non-null    int64
 1   work_year           607 non-null    int64
 2   experience_level    607 non-null    object
 3   employment_type     607 non-null    object
 4   job_title           607 non-null    object
 5   salary              607 non-null    int64
 6   salary_currency     607 non-null    object
 7   salary_in_usd       607 non-null    int64
 8   employee_residence  607 non-null    object
 9   remote_ratio        607 non-null    int64
 10  company_location    607 non-null    object
 11  company_size        607 non-null    object
dtypes: int64(5), object(7)
memory usage: 57.0+ KB
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 9   remote_ratio        607 non-null    int64
 10  company_location    607 non-null    object
 11  company_size        607 non-null    object
dtypes: int64(5), object(7)
memory usage: 57.0+ KB
       Unnamed: 0    work_year        salary  salary_in_usd  remote_ratio
count  607.000000   607.000000  6.070000e+02     607.000000     607.00000
mean   303.000000  2021.405272  3.240001e+05  112297.869852      70.92257
std    175.370085     0.692133  1.544357e+06   70957.259411      40.70913
min      0.000000  2020.000000  4.000000e+03    2859.000000       0.00000
25%    151.500000  2021.000000  7.000000e+04   62726.000000      50.00000
50%    303.000000  2022.000000  1.150000e+05  101570.000000     100.00000
75%    454.500000  2022.000000  1.650000e+05  150000.000000     100.00000
max    606.000000  2022.000000  3.040000e+07  600000.000000     100.00000
Training Score: 0.7907485429198912
Testing Score: 0.7611049686306373
Coefficient: [[0.0035702]]
Intercept: [2020.32389476]
Enter Years of Experience: 20
Estimated Salary for 20.0 years of experience is 2020.40
```

RESULT:

The Linear Regression algorithm was successfully implemented on the given dataset. The model identified the relationship between the input and output variables and predicted continuous values based on the input data.